Bug Reports — HW6: Python / pytest

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Defects found by pytest automation against the ESP32 firmware in Wokwi (24 tests: 20 passed, 4 failed). Test repository: https://github.com/alexandramoroz12/embeded_firmware-autotests_hw6

Environment: Firmware in Wokwi simulation (ESP32 DevKit), serial via rfc2217://localhost:4000; Python 3.14.7, pytest 9.1.1, pyserial

Summary

ID	Title	Severity
Bug_1	"Invalid value" error is returned after the `SET PWM 100` command	Critical
Bug_2	RESET does not restore PWM to 0 and LED to OFF	Critical
Bug_3	GET TEMP returns malformed JSON	Low/Medium
Bug_4	PING response is delayed to ~1.3 s (vs ~0.17 s for other commands), causing TimeoutError with a 1.0 s client timeout	Low/Medium

Note: Bug_4 is classified as a communication/timing issue: the functional response is correct, but the delay breaks clients that use a ~1 s timeout.

Bug_1 — "Invalid value" error is returned after the `SET PWM 100` command

Requirement	`SET PWM N` accepts every integer from 0 through 100 inclusive.
Environment	ESP32 DevKit
Preconditions	Device ACTIVE
Steps	1. Enter the `SET PWM 100` command. 2. Check the log.
Expected	{"status":"OK","pwm":100}
Actual	"Invalid value" error is returned
Raw serial evidence	`{"status":"ERROR","code":"INVALID_VALUE"}`
Pytest evidence	`FAILED tests/test_boundaries.py::test_pwm_valid_boundaries[100] - AssertionError: assert 'ERROR' == 'OK'`
Reproducibility	5/5
Severity	Critical



Figure: Raw serial evidence

Bug_2 — RESET does not restore PWM to 0 and LED to OFF

Requirement	RESET restores a safe output state: LED/PWM OFF/0.
Environment	ESP32 DevKit
Preconditions	Device ACTIVE
Steps	1. Enter the SET LED ON command. 2. Enter the RESET command. 3. Enter the STATUS command. 4. Check the log.
Expected	PWM is 0 and LED is OFF
Actual	PWM is 255, LED is ON
Raw serial evidence	`{"status":"OK","led":true,"pwm":255,"heap":350116}`
Pytest evidence	`FAILED tests/test_relay.py::test_reset_restores_safe_state - assert 255 == 0`
Reproducibility	5/5
Severity	Critical

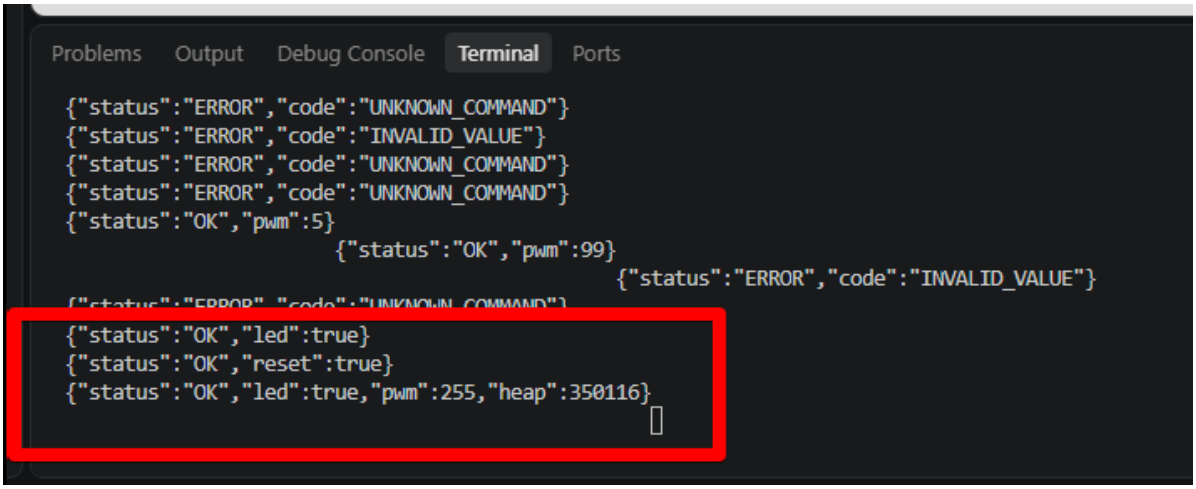


Figure: Raw serial evidence

Bug_3 — GET TEMP returns malformed JSON

Requirement	GET TEMP returns valid JSON with either a valid temperature or an explicit sensor error.
Environment	ESP32 DevKit
Preconditions	Device ACTIVE
Steps	1. Enter the GET TEMP command. 2. Check the log.
Expected	The device returns valid JSON containing either a valid temperature with status OK or an explicit sensor error with status ERROR.
Actual	The device returns malformed JSON: {"status":"OK","temperature":24.5 — the closing } is missing
Raw serial evidence	`{"status":"OK","temperature":24.5`
Pytest evidence	`FAILED tests/test_sensor.py::test_temperature_is_valid_json - AssertionError: Invalid JSON from device: '{"status":"OK","temperature":24.5'`
Reproducibility	5/5
Severity	Low/Medium



Figure: Raw serial evidence

Bug_4 — PING response is delayed to ~1.3 s (vs ~0.17 s for other commands), causing TimeoutError with a 1.0 s client timeout

Requirement	Automation must never wait forever: communication uses a finite timeout.
Environment	Firmware in Wokwi simulation (ESP32 DevKit), serial via rfc2217://localhost:4000; Python 3.14.7, pytest 9.1.1, pyserial Test harness: Device class, timeout=1.0 (set in conftest.py)
Preconditions	Device ACTIVE
Steps	1. Start the Wokwi simulation and connect over RFC2217 (115200 baud). 2. Send PING\n. 3. Measure the time until the response line arrives. 4. For comparison, send STATUS\n and measure the time. 5. Or run python -m pytest tests/test_status.py::test_ping -v with timeout=1.0.
Expected	PING responds with {"status":"OK","command":"PING"} in a time comparable to other commands (~0.2 s), i.e. within the 1.0 s client timeout.
Actual	The response is correct, but arrives after 1.28–1.34 s (6 of 6 measurements). With a 1.0 s timeout, the client raises TimeoutError: No response for command: PING. Note: PING responds with {"status":"OK","command":"PING"} when entered manually in the Wokwi Terminal.
Raw serial evidence	-
Pytest evidence	`FAILED tests/test_status.py::test_ping - TimeoutError: No response for command: PING`
Reproducibility	5/5
Severity	Low/Medium

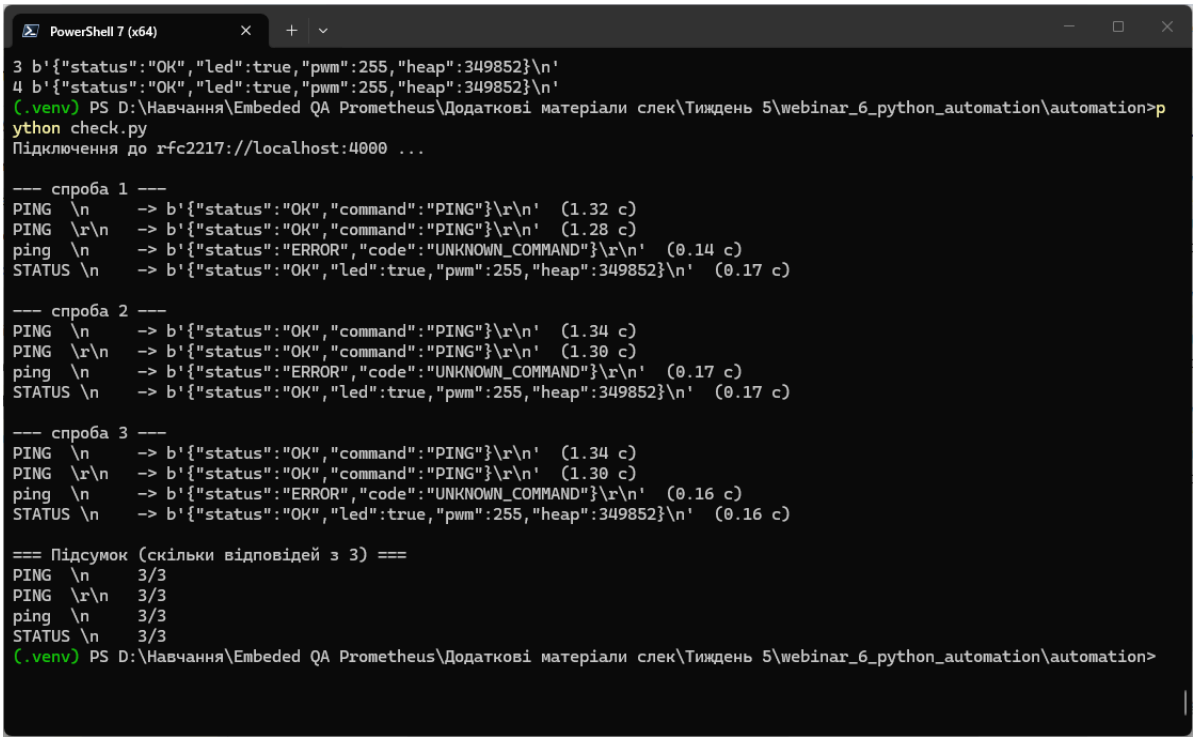


Figure: Timing measurements (PING vs STATUS)